VIGIL: Validity-Gated Retrieval for Product–Litigation Risk Entailment

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Abstract

Retrieval-augmented generation (RAG) treats semantically similar documents as interchangeable evidence, but in law a lexically close precedent may already have been narrowed, superseded, or overruled at the date that matters. We introduce Product–Litigation Risk Entailment (PLRE), a directional task that asks whether an AI product's attributes entail exposure to a legal-risk theory under precedent valid at a given query date. Our system, VIGIL (Validity-Interval Gating in Litigation), extracts treatment edges (overrules, supersedes, narrows) from later opinions, turns them into validity intervals, and validates them against gold citation spans (negative-treatment micro-F1 0.80) and a commercial citator (0.81). A simple retrieval-time gate applies these intervals before generation, and a goal-conditioned agentic loop stops when evidence coverage, not legal correctness, is satisfied. On VIGIL-Dataset (1,247 U.S. AI-litigation cases, 312 product-mapping cases), validity gating raises held-out PLRE F1 from 0.72 to 0.78 and cuts the stale-citation rate from 12.7% to 4.1% relative to the same pipeline without the gate (point estimates, not significance-tested). On a human-adjudicated triage subset ($n = 82$), VIGIL reaches macro F1 0.79 vs. 0.75 for an iteration-matched baseline. We release the dataset, label provenance, guidelines, and evaluation scripts. VIGIL is decision support for qualified reviewers, not legal advice.

1 Introduction

Generative-AI products face growing litigation over training data, output reproduction, model behavior, and downstream misuse (Henderson et al., 2023; Levendowski, 2018; Grimmelmann, 2016). High-profile suits such as *Getty Images v. Stability AI* (Getty Images, Inc., 2023), *New York Times v. Microsoft* (The New York Times Company, 2023), and *Kadrey v. Meta* (Kadrey et al., 2023) illustrate the range of theories from DMCA §1202 information removal to direct infringement and vicarious liability for outputs. For the engineering organizations that build these systems, the operational question is not merely whether a lawsuit exists, but whether a theory of liability maps onto a specific product attribute, whether the supporting authority remained valid at the relevant date, and which mitigation should be reviewed by counsel. Legal-operations teams track dockets and news, yet the bridge from a litigation event to a product-specific engineering action is usually manual. Keyword alerts surface noisy filings, docket trackers rarely map cases to product features, and conventional retrieval-augmented generation (RAG) (Lewis et al., 2020) treats semantically similar legal documents as interchangeable even when later decisions have narrowed or superseded them.

We formalize this need as Product–Litigation Risk Entailment (PLRE): deciding whether product attributes entail exposure to a legal-risk theory under time-valid precedent. PLRE differs from standard legal textual entailment (Rabelo et al., 2022) in three respects: the premise is an engineering description rather than legal text, the relation is directional (product attribute $\rightarrow$ risk theory), and the support set is conditioned on which authority was operative at the query date. Unlike commercial citators such as Shepard's and KeyCite, we construct a public-corpus, automatically extracted, time-indexed treatment layer and connect it directly to product-facing entailment.

We present VIGIL (Validity-Interval Gating in Litigation), a continuous sensing system built around this task. Its core component is a supersession-aware treatment layer that infers when each authority ceases to be operative from treatment edges in later opinions. A simple retrieval-time gate applies these intervals before generation, rather than flagging stale citations after an LLM has already produced a report. Around

this layer, a goal-conditioned agentic loop exposes unmet evidence requirements and stops when a coverage threshold is met. Although the treatment-layer construction is in principle applicable to any time-stamped corpus in which later documents can supersede earlier ones, we evaluate it only for U.S. case law within VIGIL; external-stack transfer remains a pilot and is not claimed.

Our contributions are:

- a supersession-aware treatment layer that extracts treatment edges from public opinions, validates them against gold spans and a commercial citator, and propagates them into validity intervals that gate retrieval at query time (§3.4);
- PLRE, an initial product-conditioned legal-risk entailment benchmark with non-VIGIL baselines and ablations that separate temporal validity from flat similarity (§3.5, §5);
- a goal-conditioned agentic loop with a coverage-based stopping rule; a conformal omission-risk bound holds under exchangeability and is reported as an auxiliary control (it is violated under temporal shift; §2.1, §5);
- VIGIL-Dataset with per-record provenance and a human-adjudicated subset (§4), plus a public SCOTUS overruling pair resource (see supplementary material).

2 Problem Formulation

Let $\mathcal{D} = \{d_1, \ldots, d_n\}$ be public litigation documents with metadata and text, and let P be a product specification describing architecture, data provenance, modality, and deployment context. VIGIL returns a structured risk report R^* and mitigation candidates $\mathcal{G}^*$ that must satisfy an auditable goal predicate $\mathcal{G}$ (§2.1). Report quality is not optimized directly; we evaluate it by severity-label F1 (RQ1), blinded report ratings (RQ2), and the coverage loss L_τ (RQ5). The core difficulty is a domain gap: product specifications employ engineering terms (data pipelines, API distribution), whereas litigation documents speak in causes of action, procedural posture, and precedent. PLRE (§3.5) therefore requires mapping engineering-language attributes onto legal-risk theories while conditioning the support set on authority that remains valid at the query date.

2.1 Goal Predicate and Stopping Rule

VIGIL treats report generation as goal-conditioned planning. The predicate is

$$\mathcal{G}(R) = \mathcal{G}_{\text{risk}}(R) \wedge \mathcal{G}_{\text{evidence}}(R) \wedge \mathcal{G}_{\text{guide}}(R), \quad (1)$$

where $\mathcal{G}_{\text{risk}}$ requires every risk item to carry a severity label, $\mathcal{G}_{\text{evidence}}$ requires at least k supporting documents above relevance threshold τ, and $\mathcal{G}_{\text{guide}}$ requires mitigation text for each High or Medium item. The controller selects among SEARCH, ANALYZE, SUMMARIZE, MAP, VERIFY, and REPORT; the deterministic (LLM-free) Goal Checker either terminates, emits a structured critique that closes residual gaps, or escalates unmet conditions.

Let $\Phi(R_t)$ count the unmet atomic sub-conditions of $\mathcal{G}$, and let $\mathcal{T}_c^H$ be the human-labeled High-risk theories of case c that are retrievable—i.e., supported by at least one authority in $\mathcal{D}$ that is valid at t_q. The coverage loss at evidence threshold τ is

$$L_\tau(c) = \mathbf{1}\left[\exists\, T \in \mathcal{T}_c^H : T \notin R^*\right]. \quad (2)$$

Observation 1 (Coverage-controlled stopping with escalation)**.** *Assume the set of risk items under consideration is fixed within a run (the Reporter may rewrite mitigation text but does not introduce new risk atoms that expand Φ). Under monotone evidence caching and attainable progress, $\Phi(R_t)$ is non-increasing, reaches zero in at most $\Phi(R_0)$ successful progress steps when all atoms are attainable, and otherwise returns the residual atoms at $T_{\max}$. If $\hat{\tau}$ is selected on exchangeable calibration cases by conformal risk control, then the next-case omission risk satisfies $\mathbb{E}[L_{\hat{\tau}}] \leq \alpha$.*

The first clause follows almost directly from the definitions. The second clause is a standard application of conformal risk control (Angelopoulos et al., 2024) to L_τ, which is monotone in τ. It governs evidence coverage only, holds only for exchangeable cases—our temporal split violates it at $\alpha \in \{0.05, 0.10\}$ (§5)—and its tightness depends on the calibration-set size (150 cases).

3 Method: The VIGIL Framework

3.1 Overview

VIGIL comprises two layers. A sensing pipeline discovers AI-litigation filings in CourtListener/RECAP (Free Law Project, 2023) and legal news and converts them into case records; this engineering infrastructure is documented in the

supplementary material. The remainder of this section describes the evaluated NLP components: LegalRisk-RAG (§3.2), the goal-conditioned agentic loop (§3.3), the temporal precedent graph (§3.4), and the PLRE decision procedure (§3.5).

3.2 LegalRisk-RAG

The LegalRisk-RAG pipeline adapts standard RAG to legal text in three respects. **Chunking** segments documents at logical structural boundaries—numbered claims, "WHEREFORE" clauses, section headings—rather than fixed token windows; each chunk retains a metadata header (case identifier, filing date, court level, section type). **Embedding:** we compared a dense retriever (text-embedding-3-large), RECAP-tuned LegalBERT (Chalkidis et al., 2020), and a hybrid BM25 (Robertson and Zaragoza, 2009) + neural re-ranker on a 200-pair development set; the hybrid performed best and is the default. **Dynamic context expansion (precedent chaining):** when a retrieved chunk references a prior case, the pipeline appends that opinion as secondary context and queries the temporal precedent graph for subsequent treatment, so that the Analyzer receives both semantic neighbors and validity metadata.

3.3 Goal-Conditioned Agentic Loop

The four specialized agents and the central Goal Checker operate as described in the supplementary material. The Planner decomposes unmet sub-goals into ordered retrieval/analysis subtasks; the Retriever runs LegalRisk-RAG queries and reformulates via legal entity term expansion when $\mathcal{G}_{\text{evidence}}$ is unmet; the Analyzer computes $\text{RiskScore}(d_i, P, C)$, clusters evidence, and maps clusters to a risk taxonomy (Training Data, Output Reproduction, API Misuse, Model Weights); the Goal Checker evaluates $\mathcal{G}(R_t)$ after each pass and returns machine-readable residuals; the Planner converts these critiques into expanded entity/date/theory queries; and the Reporter assembles the risk register, per-risk evidence with citations, and mitigation guidance.

3.4 Temporal Precedent Graph

VIGIL's central component is a treatment layer that determines when each authority ceases to be operative; a simple retrieval-time gate then applies it. The layer is realized as a temporal precedent graph $P_t = (V, E_t)$. Each node $v \in V$ denotes a case, order, or doctrinal rule and stores jurisdiction, court level, issue tags, filing and decision dates, and a validity interval $[s_v, e_v)$, where e_v remains open unless the node is later vacated, overruled, or superseded; material narrowing instead lowers its treatment weight. Directed edges encode treatment relations: CITES, DISTINGUISHES, NARROWS, SUPERSEDES, and OVERRULES. Retrieval for a query at date t_q first obtains semantically relevant candidates and then applies validity gating:

$$\text{score}(v, q, t_q) = \text{sim}(v, q) \cdot \mathbf{1}[s_v \le t_q < e_v] \cdot w(\text{treat}(v, t_q)). \quad (3)$$

Here $w(\cdot)$ is a fixed monotone schedule: positively treated or still-open authorities receive weight 1.0; cautionary authorities receive 0.75; material NARROWS receives 0.50; fully closed (overruled or superseded) authorities are already excluded by the indicator. The operator separates legal authority from topical similarity: a stale but lexically close case may still appear as context, yet it cannot support a risk decision without passing the validity check. Equation (3) is therefore a constrained retrieval operator over (q, t_q): a host retriever scores semantic relevance as usual, and PLRE receives only support whose legal status remains operative at t_q.

Reference	P	R	F1
Gold spans ($n = 600$)	0.84	0.77	0.80
Citator (Protocol P2)	0.83	0.80	0.81

Table 1: Treatment-layer validation (neg.-treatment micro over OVERRULES/SUPERSEDES/material NARROWS).

Why not a date filter? Once intervals are known, Eq. (3) is a metadata filter any vector store can apply. The difficulty is upstream and conceptual: e_v is set by later documents that overrule, supersede, or narrow v, so it is not recoverable from v's own filing date. A pure recency or document-date filter therefore cannot implement the same operator; the empirical question is whether treatment edges can be extracted reliably enough to use. Table 1 addresses that question: negative-treatment micro-F1 is 0.80 on 600 gold spans and 0.81 against a commercial citator (Protocol P2; missed-action rate 20.4%; 3.8% of held-out PLRE pairs flip after citator correction).

Graph construction. Nodes are induced from CourtListener/RECAP with intervals initialized to

$[s_v, \infty)$. Edges are extracted by a citation-span detector plus an LLM classifier (GPT-4o, $T = 0$); interval-closing edges must be anchored to a detected span, decision date, and source document, and gold evaluation uses spans held out from prompts. A confirmed OVERRULES/SUPERSEDES edge at t sets $e_v \leftarrow t$; material NARROWS sets $w = 0.50$ from t. The graph has 1,812 nodes and 4,760 edges. Low-confidence negative-treatment edges are routed to human review rather than auto-closing intervals.

3.5 PLRE Decision Procedure

We cast product-litigation alignment as a risk-entailment task. Given a product spec P with attributes $\{p_1, \ldots, p_m\}$, a legal-risk theory T_j, and a query date t_q, PLRE predicts

$$y_{ij} = \mathbf{1}[P, p_i, P_{t_q} \models T_j]. \tag{4}$$

Unlike symmetric similarity, PLRE is directional. A pair (p_i, T_j) is promoted to the risk register only when (i) the entailment score exceeds the validation-set threshold, (ii) the top supporting precedents remain valid at t_q, and (iii) the report includes mitigation text grounded in at least three cited authorities when available.

PLRE gold labels are constructed from anonymized product-attribute cards and docket-grounded theory cards. Annotators judge a directional pair as positive only when the product attribute, legal theory, query date, and at least one valid supporting authority jointly establish exposure; negative pairs comprise ordinary mismatches and stale-authority distractors. Critically, the validity judgment encoded in the labels is produced by annotators reading docket and opinion text, independently of VIGIL's automatically extracted treatment graph, so the gate must recover label validity rather than presuppose it. Product archetypes are tracked in the split manifest, and year and litigation type are stratified to reduce temporal and topical leakage. The formulation is system-agnostic: PLRE is directional entailment subject to an exogenous, time-indexed support constraint.

4 Dataset

VIGIL-Dataset is a curated benchmark of AI-related copyright and intellectual-property litigation designed for structured risk analysis.

Attribute	Value
Total cases	1,247
Total documents	8,934
Coverage period	2020–2025
Unique defendants	43
Risk labels	3 (High/Med/Low)
Status: Initial / Active	298 / 695
Status: Settled / Dismissed	148 / 106
Product-mapping GT cases	312

Table 2: VIGIL-Dataset summary statistics. Status counts sum to 1,247.

4.1 Construction and Statistics

The dataset compiles CourtListener/RECAP filings and supplementary legal-news signals spanning January 2020 to December 2025, with the majority of filings concentrated in 2023–2025. Cases were obtained via a retrospective keyword search of RECAP complaints and were manually screened to confirm that AI development, training, or deployment was at issue. The scope is intentionally broader than copyright-only trackers: it includes privacy, publicity, product-liability, and contract theories when AI systems are material to the dispute, and counts each docket separately (including related and consolidated filings). One case corresponds to one docket; consolidated actions are therefore counted per docket. Table 2 summarizes the final annotated subset (1,247 dockets after screening and dedup; ≈980 distinct disputes). Public AI-copyright trackers typically list a few hundred distinct disputes. Our 1,247 figure counts dockets, not disputes: consolidated and related actions are retained as separate records. After merging on dispute identity the unique-dispute count is approximately 980. The top-three defendants account for 41.3% of dockets, reflecting the concentration of high-profile generative-AI litigation; we release the defendant-frequency table so that users can assess applicability to smaller vendors.

4.2 Annotation and Availability

An automated pre-labeler first assigns a triage score using an additive scoring matrix (supplementary material). Three legal research specialists then independently review a stratified sample of 400 cases under a fixed guideline. During adjudication, annotators see docket text, product attributes, candidate theory cards, and query dates, but neither system identity nor the pre-labeler's score. The remaining 847 cases retain automated pre-labels; a 5% spot-check yields 91% agreement with at least one

specialist. Inter-annotator agreement on the 400-case sample is mean pairwise Cohen's $\kappa = 0.74$ (Fleiss' $\kappa = 0.72$). Because retained labels can inflate metrics, we release per-record provenance, restrict all primary claims to the human-only evaluation, and treat retained-label results solely as scale-oriented comparisons. The dataset, guidelines, split files, seeds, and evaluation scripts are supplied as anonymized supplementary material.

5 Experiments

We evaluate VIGIL along five research questions:

- **RQ1:** Does VIGIL improve severity-label agreement over strong baselines?
- **RQ2:** Do blinded specialists rate VIGIL reports more highly, especially on evidence sufficiency?
- **RQ3:** How much do the temporal gate and agentic loop each contribute?
- **RQ4:** Does validity gating improve Product–Litigation Risk Entailment?
- **RQ5:** Does the calibrated stopping rule control evidence-omission risk under temporal shift?

5.1 Experimental Setup

Dataset split and label provenance. VIGIL-Dataset is partitioned into stratified train/validation/test splits by year and litigation type. Because a substantial share of labels originate from an automated pre-labeler and only a subset are human-adjudicated, primary claims are computed exclusively on the human-adjudicated test split ($n = 82$ cases); retained-label results are reported solely for scale. Conformal calibration draws on human-adjudicated cases from the training and validation pools, kept disjoint from the test cases. For VIGIL and baselines B2–B5 the training split is used only to select few-shot examples and tune thresholds; no LLM is fine-tuned.

PLRE pairs (1,856 total; 28.4% positive; 19.7% stale-authority distractors) are partitioned 1,112 / 372 / 372 (train / validation / test) by product archetype and year, corresponding to an approximate 60/20/20 ratio. Few-shot exemplars and the entailment threshold use only train/validation; all reported metrics use the 372 held-out test pairs.

System	Human-adj.			Full
	F1	P	R	F1
B1 Additive-only	0.55	0.55	0.56	0.62
B2 Single-Pass LLM	0.70	0.70	0.71	0.72
B3 RAG-only	0.72	0.72	0.72	0.74
B4 Single-Agent	0.74	0.74	0.74	0.76
B5 Self-Refinement	0.75	0.75	0.75	0.77
VIGIL Full	**0.79**	**0.79**	**0.80**	**0.81**

Table 3: Macro severity-label agreement. Human-adjudicated subset ($n = 82$; primary). Full: held-out test including retained pre-labels.

Baselines. We compare VIGIL against five risk-classification baselines: B1 Additive-only Rule; B2 Single-Pass LLM; B3 RAG-only; B4 Single-Agent (VIGIL without the goal-conditioned loop); and B5 Iteration-Matched Self-Refinement (Reflexion-style (Shinn et al., 2023), matched on VIGIL's mean budget of 2.4 iterations). B2–B5 share the same GPT-4o backbone and decoding settings as VIGIL. For PLRE we further include non-VIGIL baselines that isolate suspected confounds: product–theory cosine, LegalBERT product–theory NLI, COLIEE-style (Rabelo et al., 2022) precedent retrieval without interval closure, and VIGIL without temporal gating.

Automatic evaluation. All reported automatic metrics freeze the treatment graph produced by the extractor; low-confidence edges are not human-reviewed at evaluation time. We report macro F1/precision/recall over three severity labels, report-quality ratings by two blinded specialists, loop efficiency, PLRE precision/recall/F1, edge-error propagation, and evidence-coverage omission risk under calibrated thresholds. Severity-label F1 measures triage-label agreement, not legal liability.

5.2 RQ1: Severity-Label Agreement

Table 3 reports severity-label agreement on the human-adjudicated subset ($n = 82$). VIGIL reaches 0.79 macro F1 vs. 0.75 for iteration-matched B5; the same rank order holds on the full held-out test set. These triage numbers are secondary to PLRE and are not significance-tested.

5.3 RQ2: Report Quality

Two blinded specialists rated VIGIL and B2–B5 reports on 15 cases (75 reports) along Accuracy, Completeness, Actionability, Evidence Sufficiency, and Clarity (5-point scales). Mean scores were highest for VIGIL on Evidence Sufficiency (4.4).

Variant	F1	ΔF1	Stale %
VIGIL Full (gate + weight)	0.78	—	4.1
− treatment weighting $w(\cdot)$	0.75	−0.03	7.8
− interval gating (flat sim.)	0.72	−0.06	12.7

Table 4: Temporal precedent-graph ablation (main claim). Both mechanisms of Eq. (3) are removed in sequence; the bottom row recovers flat semantic similarity. F1 is PLRE F1; Stale is the stale-citation rate, i.e., citations to already-closed precedent ($t_q \geq e_v$).

System	P	R	F1
Product–theory cosine	0.64	0.66	0.65
LegalBERT NLI	0.69	0.68	0.68
COLIEE-style, no gate	0.71	0.70	0.70
RAG-only generation	0.73	0.74	0.73
VIGIL w/o gating	0.72	0.73	0.72
VIGIL Full	**0.77**	**0.80**	**0.78**

Table 5: PLRE comparison against external legal-NLI and precedent-retrieval baselines (LegalBERT NLI: product–theory NLI). Held-out test pairs only.

Mean inter-rater $\kappa = 0.68$. RQ2 is supporting evidence only ($n = 15$); we report system means in the supplementary material and treat pairwise tests as exploratory.

5.4 RQ3: Agentic Loop and Temporal Graph Ablations

Temporal precedent reasoning is the central empirical claim. Table 4 isolates the two mechanisms of Eq. (3). Full validity-interval gating plus treatment weighting yields PLRE F1 of 0.78 and a stale-citation rate of 4.1%. Removing treatment weighting alone drops F1 by 0.03 and roughly doubles stale citations (7.8%). Removing interval gating entirely (recovering flat semantic similarity) widens the gap to $\Delta\text{F1} = -0.06$ and raises the stale-citation rate to 12.7%—a three-fold increase. Flat retrieval systematically promotes lexically similar but already-closed authorities over still-operative ones, altering the evidence that reaches the generator. This controlled degradation constitutes the paper's main methodological result.

Negative-treatment error propagation was measured by substituting gold spans and by additional masking of extracted edges. The extractor row operates at the observed miss rate (≈23%, from 0.77 recall) and costs 2 PLRE F1 points relative to gold-span substitution. Additional masking of extracted negative-treatment edges further degrades performance, confirming that the gate's benefit depends on treatment-edge quality (full numbers in the supplementary material).

5.5 RQ4: Product–Litigation Risk Entailment

PLRE uses anonymized product-attribute cards (five archetypes) paired with docket-grounded theory cards across 312 product-mapping cases (1,856 pairs; 28.4% positive; 19.7% stale-authority distractors; $\kappa = 0.71$). Table 5: VIGIL Full reaches 0.78 F1 (+0.06 vs. no-gate; +0.05–0.13 versus non-VIGIL product–theory and legal-NLI baselines). VIGIL without gating (0.72) sits slightly below RAG-only (0.73), indicating that the agentic loop without a validity filter can reinforce stale but lexically close precedent; the gate is the component that improves PLRE. Gains partly reflect rejection of stale distractors (19.7% of pairs); a non-distractor subset analysis is provided in the released evaluation scripts to address circularity concerns. Primary claims rest on the 372 held-out pairs.

5.6 RQ5: Evidence-Coverage Risk Control and Temporal Shift

Calibrating the evidence threshold to $\alpha \in \{0.05, 0.10, 0.20\}$ yields omission risk 0.043/0.082/0.171 in random splits. Under a temporal split (2023–24 → 2025) the achieved risk rises to 0.061/0.113/0.196, exceeding the target at $\alpha = 0.05$ and 0.10, which motivates rolling re-calibration. An open-weights Llama-3.1-70B (Grattafiori et al., 2024) + bge (Xiao et al., 2024) stack runs unchanged, with full-test-set macro F1 0.81 → 0.76 and 2.4 vs. 2.6 iterations.

6 Related Work

Legal NLP and domain models. Legal NLP has studied judgment prediction (Chalkidis et al., 2019), statutory reasoning (Holzenberger et al., 2020), legal benchmarks such as LexGLUE (Chalkidis et al., 2022) and LegalBench (Guha et al., 2023), case retrieval and entailment (Rabelo et al., 2022), and summarization (Huang et al., 2021; Manor and Li, 2019), often with domain-adapted encoders including LEGAL-BERT (Chalkidis et al., 2020), CaseHOLD (Zheng et al., 2021), and Lawformer (Xiao et al., 2021). Audits show that LLMs and commercial legal RAG tools still hallucinate (Dahl et al., 2024; Magesh et al., 2025). Prior work also studies overruling detection as a classification task over case text (Zheng et al., 2021); we instead extract

treatment edges to build time-indexed validity intervals and apply them as a retrieval gate. These lines of work take legal text as the premise, treat the corpus as static, and do not test whether retrieved authority remained operative at a given date. PLRE changes the inference unit: product attributes are mapped to legal-risk theories in a directional, product-conditioned, and temporal manner.

Time-sensitive QA and temporal IR. Time-sensitive QA conditions answers on the moment a question is asked (Chen et al., 2021; Zhang and Choi, 2021; Liška et al., 2022; Kasai et al., 2023); temporal IR uses document time as a ranking signal (Campos et al., 2014; Dhingra et al., 2022). Both treat time as a property of facts or queries. In PLRE, time enters through supersession: a later document alters the evidential status of an earlier one. Commercial citators such as Shepard's (LexisNexis, 2024) and KeyCite (Thomson Reuters, 2024) track negative treatment; citation-network analysis and edge labeling characterize the underlying graph (Fowler et al., 2007; Sadeghian et al., 2018). VIGIL attaches validity intervals to extracted treatment edges and applies them as a retrieval-time gate.

RAG, knowledge conflicts, and agents. RAG typically executes a fixed retrieval plan before generation (Lewis et al., 2020; Karpukhin et al., 2020; Gao et al., 2023); work on knowledge conflicts and outdated parametric knowledge shows that generators may prefer stale but familiar evidence unless retrieval is constrained (Longpre et al., 2021; Chen et al., 2022; Xie et al., 2024). Agent frameworks benefit from iterative tool use yet stop when the model itself deems its answer satisfactory (Yao et al., 2023; Shinn et al., 2023; Wei et al., 2022; Park et al., 2023; Wang et al., 2024a,b). VIGIL ties stopping to an externally inspectable predicate over evidence support and mitigation coverage, and employs conformal risk control (Angelopoulos et al., 2024, 2025; Geifman and El-Yaniv, 2017) narrowly for evidence omission rather than legal correctness.

Foundation models, copyright, and fair use. Foundation-model training and output raise copyright and fair-use questions that motivate product-facing risk mapping (Henderson et al., 2023; Levendowski, 2018; Grimmelmann, 2016; Bommasani et al., 2021). High-profile suits such as *Getty Images v. Stability AI* (Getty Images, Inc., 2023), *New York Times v. Microsoft* (The New York Times Company, 2023), and *Kadrey v. Meta* (Kadrey et al., 2023) illustrate the range of theories from DMCA §1202 to direct infringement and vicarious liability. Infrastructure resources such as CourtListener (Free Law Project, 2023) and the Caselaw Access Project (Harvard Law School Library, 2024) enable public-corpus construction of treatment layers that commercial citators already maintain privately.

7 Discussion

The central empirical result is that a validated treatment layer applied as a hard retrieval gate changes PLRE behavior under matched conditions: stale citations fall from 12.7% to 4.1% and F1 rises by 0.06 relative to the identical pipeline without gating (Table 4). Independent checks place negative-treatment micro-F1 at 0.80 (gold spans) and 0.81 (commercial citator). These figures are point estimates; we treat them as evidence for the mechanism rather than as significance-tested superiority. Instance-level predictions are retained and released so that paired bootstrap CIs can be attached by any reader.

Two controls bound the claim. First, without the gate the agentic loop alone does not improve PLRE (0.72 vs. RAG-only 0.73), so gains are not explained by extra test-time compute. Second, a surface-form heuristic recovers many explicit OVERRULES/SUPERSEDES mentions but almost no material narrowing, and citator-oracle intervals flip only 3.8% of held-out pairs—the extracted layer already tracks the commercial reference for downstream decisions. We deliberately do not evaluate pure date filters or learned temporal re-rankers on PLRE in the present study; a date filter cannot recover supersession endpoints from later opinions, which is the extraction problem this work targets. External-stack transfer (Protocol P1) remains a pilot and is not claimed.

On severity triage ($n = 82$), VIGIL leads iteration-matched B5 by a small margin (0.79 vs. 0.75); that comparison is secondary to PLRE.

8 Conclusion

We introduced PLRE and VIGIL, whose extracted treatment layer gates retrieval by operative authority. On held-out pairs, gating lifts PLRE F1 by 0.06 and cuts stale citations from 12.7% to 4.1% relative to flat similarity. The layer reaches negative-treatment micro-F1 0.80 on gold spans and 0.81

against a commercial citator.

Limitations

What this paper does and does not claim. Our primary claim is mechanistic: an extracted treatment layer, validated against gold spans and a commercial citator (negative-treatment micro-F1 0.80 / 0.81), can be applied as a hard validity gate at retrieval time. Doing so changes PLRE outcomes and stale-citation rates relative to the identical pipeline without the gate (Table 4: F1 $0.72 \rightarrow 0.78$; stale-cite $12.7\% \rightarrow 4.1\%$). We report point estimates and do not claim statistical significance for any comparison. The contribution is the validated gate construction and its downstream effect under matched retrieval, not a leaderboard win under every possible baseline.

Statistical uncertainty. Instance-level predictions necessary for the main automatic metrics (stale-citation rate, decision-flip under citator correction, masking ablations) were retained and are released with the evaluation scripts. Paired bootstrap confidence intervals on the primary PLRE comparison (gate vs. no-gate) can therefore be recomputed by any reader. In the present submission we report point estimates together with the operational magnitudes that matter for the mechanistic claim. Aggregate mean$\pm$std over five random seeds appear in the supplementary material as run-level summaries.

Baselines not evaluated on PLRE. B5 (iteration-matched self-refinement), token-matched refinement, pure document-date filters, and learned temporal re-rankers were not run on the PLRE task. A pure date filter cannot encode supersession by later opinions (the endpoint e_v is not a property of document v), so it is not a substitute for treatment extraction; we leave a head-to-head comparison to follow-on work. B5 on severity labels is a triage control only. The no-gate ablation already isolates the contribution of the validity indicator under otherwise identical retrieval and generation.

Distractor composition. Stale-authority distractors constitute 19.7% of PLRE pairs by design: they test whether the gate removes closed support that lexical retrieval would otherwise promote. Gains partly reflect rejection of these distractors; a non-distractor subset analysis is left to the released evaluation scripts and future work to fully isolate ranking among still-open authorities.

Split structure. Case and PLRE splits are stratified by year and litigation type (or product archetype), not by dispute identity. Related dockets of one dispute can therefore cross the train/test boundary. Primary PLRE metrics use held-out pairs only, with few-shot exemplars and thresholds taken from train/validation, but a dispute-blocked re-split remains desirable and is listed as future work. With only five product archetypes, leave-one-archetype-out evaluation would further stress generalization and is also left to future work.

Extractor quality, transfer, and raters. Negative-treatment recall is 0.77 on gold spans (citator missed-action rate 20.4%). Protocol P1 (external-stack transfer of the gating modifier) is reported only as a pilot; the pilot does not meet the pre-specified acceptance criterion and we therefore make no transfer claim. Annotators and report raters were institutional staff; external replications can use the released guidelines. Conformal coverage weakens under temporal shift, motivating rolling recalibration in deployment. The dataset is U.S.-federal and skewed toward large defendants (top three account for 41.3% of cases).

Reproducibility. Primary experimental runs were conducted after the December 2025 cutoff of the dataset window. The treatment graph, PLRE labels, and all automatic metrics were computed on the frozen corpus that ends 31 December 2025. Controller and Analyzer calls used the OpenAI API snapshot `gpt-4o-2024-08-06`. That version string has since been retired by Azure OpenAI; bit-identical re-runs therefore require the original snapshot or an equivalent open-weights stack (we provide a Llama-3.1-70B + bge robustness result). Because many underlying opinions pre-date the knowledge cutoff of the GPT-4o snapshot, we cannot rule out that the model had already seen portions of the corpus during pre-training; the open-weights run provides a partial check against this confound.

Future work. Larger human adjudication, paired-bootstrap CIs on the main tables, non-distractor and dispute-blocked audits, B5/token-matched and temporal re-ranker controls on PLRE, leave-one-archetype-out evaluation, and broader jurisdictional coverage.

Ethical Considerations

Decision-support boundary. VIGIL emits risk intelligence, evidence summaries, and mitigation candidates with provenance and residual uncertainty for qualified human reviewers. It is not legal advice: it does not replace counsel, predict litigation outcomes, or recommend business decisions, and severity labels are triage proxies that are not interpreted as liability. Every High- or Medium-risk item is presented for review by a qualified person, and unresolved legal-status atoms are surfaced as such rather than resolved automatically.

Human oversight of failure modes. Low-confidence negative-treatment edges enter a human review queue, since missed negative-treatment actions are the failure mode most likely to admit stale authority into a report. Reviewers can distinguish a genuine product–theory match from a retrieval gap or an unresolved legal-status ambiguity.

Data governance. Released data are public docket metadata and machine-generated summaries derived from CourtListener/RECAP; we do not redistribute attorney-authored complaint text beyond what is already public, and the P2 citator study releases only the cross-walk, sampling seeds, and agreement statistics, with no redistributed proprietary citator content. The released dataset is distributed under a CC-BY-4.0 license for the machine-generated annotations and metadata; underlying docket text remains subject to the original CourtListener/RECAP terms of use. Protocol P2 used a commercial citator under the authors' institutional research subscription. Because the precise product name is omitted for licensing reasons, exact reproduction of the P2 numbers requires an equivalent institutional subscription. We release the mapping rules so that any subscriber can re-run the agreement calculation.

Annotators and raters. The three legal research specialists who adjudicated labels and the two specialists who rated reports are staff of the authors' institution and performed this work within their regular paid duties. Before annotation began, participants received a written description of the task, the intended research use of their labels, data-handling practices, and the voluntary nature of participation; verbal consent was obtained and recorded. The work consisted solely of labeling public court documents and collected no personal information about the participants beyond institutional employment status. Under the authors' institutional human-subjects policy for secondary analysis of public records with no identifiable private information about research subjects, the activity was determined not to constitute human-subjects research requiring IRB review. This determination is recorded in the released annotation guidelines. Institutional affiliation of annotators is a potential source of bias (Limitations). Annotation guidelines are released with the dataset so that external replications can use independent raters.

Use of AI assistants. AI writing assistants were used to edit and polish the text of this paper. An AI coding assistant also wrote the code for the P1 pilot, following a design fixed by the authors; the pilot design was frozen before any run, and its results were reported as produced. Research ideas, system design, and the remaining experiments and analyses are the authors' own.

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A Additional Material

Full appendices—including dataset inclusion criteria, ingestion scoring matrix, system architecture, treatment-edge extraction accuracy, additional ablations, hyperparameters, proof of Observation 1, validation protocols P1/P2, and the SCOTUS overruling pair resource—are provided in the supplementary material uploaded with this submission. Reviewers are not required to consult the supplementary material for the claims in the main paper.